

NISHIL PATEL

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Professional Summary

AI Engineer experienced in building and deploying intelligent backend systems, retrieval pipelines and agentic workflows. Skilled in Python, FastAPI, LangGraph and multiple LLM orchestration with a focus on reliable data grounding, latency optimization, evaluation and robust failure handling.

Technical Skills

- Programming:** Python, Typescript, SQL
- Generative AI:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Context Engineering
- Agentic AI:** LangChain, LangGraph, LangSmith, CrewAI, Multi-Agent Systems, Tool Calling
- Machine Learning:** Scikit-learn, Hugging Face, TensorFlow, Keras, Pandas, NumPy, Feature Engineering, Model Evaluation
- LLMOps:** Prompt Versioning, LLM Evaluation, Token/Cost Optimization, Caching, Latency Optimization
- Backend & Databases:** FastAPI, Flask, REST APIs, Pydantic, PostgreSQL, Vector DBs, ChromaDB
- Cloud & DevOps:** Microsoft Azure, Docker, Apache Airflow, Git, GitHub

Professional Experience

Fortifai Inc.

AI Engineer

July 2025 – Present

Ahmedabad, India

- Engineered and maintained backend services for **D7**, a production due diligence platform, integrating model orchestration, data processing and report generation workflows into scalable API services.
- Owned the design and implementation of configurable **Quality Control (QC) workflows**, establishing automated validation across generated reports to improve consistency, data integrity and release reliability.
- Architected a document compliance workflow using **LangChain and LangGraph** and collaborated with the Data Engineering team to integrate **Apache Airflow** orchestration supporting enterprise AI data pipelines

Alchemyte Data Solutions

Data Science Engineer Intern

January 2025 – June 2025

Ahmedabad, India

- Engineered a natural language-to-SQL generation workflow for audit teams, translating business questions into executable SQL across complex **PostgreSQL** procurement schemas.
- Designed and refined the query-generation layer using **Python, Vanna AI and GPT-4**, improving query reliability through schema-aware prompting, validation rules and controlled SQL generation.

Projects

D7 - 3rd Party Due Diligence Platform

Python, FastAPI, PostgreSQL, Claude, OpenAI

- Engineered a data-to-report pipeline processing **70+ due diligence reports** across varying risk dimensions, transforming structured analytical data into dimension-level narratives, risk assessments and consolidated reports.
- Reduced token consumption by approximately **33%** through structured JSON chunking, token streaming and prompt caching, while reducing report-generation latency by **33%** (45 minutes to 30 minutes).
- Built a multi-layer QC framework validating **100+ checks per report** across narratives, numerical values and risk scores against source data, reducing manual validation effort by approximately **70%**.

Enterprise AI Compliance Assistant

Python, FastAPI, OpenAI, LangGraph, pgvector

- Architected a **4-agent** LangGraph workflow for enterprise document retrieval, risk analysis, compliance report generation and output validation.
- Evaluated agent-based document processing against Python PDF extraction, achieving more reliable handling of complex document structures, tables, relevant information and multiple document formats.
- Implemented multi-layer output validation using source-document cross-checking, structured schema validation, hallucination and grounding checks; reduced workflow latency by **50%** (8 minutes to 4 minutes) through context engineering.

Education

Pandit Deendayal Energy University (PDEU) Gandhinagar, India

B.Tech. in Computer Engineering

2021 – 2025

CGPA: 8.67/10